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ORIGINAL RESEARCH

Classification of fragments: recognition of artistic style

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Abstract
Disruptive phenomena are known for damaging the original appearance of artworks, even when the pieces are carefully and meticulously put together. Machine Learning and Computer vision techniques may be effective support tools during the works in an excavation site to resolve the issues of reconstructing frescoes, painting or sculptures. Not infrequently happens that multiple and different pictorial styles may be found in the same historical ruins. Classifying the fragments into the right pictorial style plays a significant role as a prior step for reconstruction. Determining the pictorial style or the painter from the full artwork is feasible, but it is distant from the real working conditions in the excavations. In this work a dataset called CLEOPATRA is proposed. It collects frescoes of eleven painting styles, from Prehistory to Surrealism, and simulate the effects of collapsing events that turn them into pieces. The fragmentation strategies applied are varied in terms of size (each work is divided into 10, 20, 40 and 80 pieces respectively) and the modus operandi for obtaining the pieces (one approach exploits the salient features of the work while another is random). For each of the fragments obtained, two different Machine Learning approaches were involved to solve the classification problem, discussing the advantages and critical aspect that make it a challenging but also relevant problem for archaeologists. The approach that exploits information on colour, shape and texture features is the one that gives the best results. A further study in which starting with the three most disparate artistic periods and progressively adding two more shows clearly that when the problem is significantly simpler, i.e. with fewer classes, the overall performance of the system improves.

Keywords Machine learning · Image fragmentation · Classification · Image processing · Image reconstruction

1 Introduction

Digital solutions applied to artistic sector can solve a broad collection of issues (Dondi et al. 2021). Computer scientists have studied for many years problems correlated to this field, proposing various approaches to help experts and restorers in their work (Sizyakin et al. 2022). Systems that could identify the strokes, the style, the art movement are examples of tasks that can be difficult to generalise among several pictorial styles even if some tries have been done in this direction (Willats and Durand 2005; Li and Wang 2004). The analysis of the visual features, like traits, lines, curves, colors and so on, can reveal hidden details of the original work and then enable the classification of the historical period even in case of severely damaged finds (Peng and Chen 2015). Inspired to the usual operative conditions for archaeologists when working in the excavation we looked for a solution for the classification of the historic artistic period of digital artefacts from their fragments. We combined the traditional humanistic perspective with computer science techniques. This work reveals how challenging the problem can be even for the most famous and performing Machine Learning approaches. Can fragments still include sufficient information to classify a painting into a certain style? To such a purpose we first created a dataset that simulated the working conditions of archaeologists and then analyzing different approaches to obtain benchmarks to deal with it. Rather than addressing

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the issues of matching the pieces together like in a puzzle, we investigated on how effectively Machine Learning solutions were useful to solve the classification problem. The proposed dataset is called CLEOPATRA (CLassification of frEscOes PictoriAl sTyle fRom frAgments). It includes eleven different art-historical periods and a minimum of 40 images have been collected per period, which are then fragmented into pieces. The process of fragmentation has been set in two modalities: a random approach and a salience-based approach, taking advantage of two well-known techniques (SLIC and SIFT, respectively) used in the state of the art in a wide variety of work (Sewan and Altaei 2021). SIFT is merely a way to extract features from fragments useful for producing a fragmentation that ensures that each fragment contains enough features to analyse it without these necessarily being semantically connected, so as to produce a fragmentation that is similar to the realist fragmentation in which the pieces are formed by falling/breaking, but avoids producing fragments that are impossible to process due to the total absence of features (e.g. a piece of sky or, more generally, a monochromatic area of the fresco devoid of any cue for analysis). Approaches that are more current and effective (Sethi et al. 2018), functioning at a very fine grain, may result in fragmentation that is unrealistic, as each fragment will contain features linked by semantic content, thereby simplifying the problem under examination. The differences about the two fragmentation modalities are better explained in the following sections. They are meant to propose two scenarios with different level of complexity. In the random fragmentation, the images of the frescoes are turned into pieces by a random criterion. In the salience-based approach, the pieces are obtained from the original image so that each of them owns a sufficient amount of visual features to be more easily classified. The pieces have been used to train Deep and Machine Learning algorithms, by using just the ones from random, the salience-based and both modalities of fragmentation. Given the freshness of the subject of fresco fragment stylistic classification, we believe that our work can provide researchers with a helpful starting point. The rest of the paper is organised as follows: Sect. 2 provides methodological assumption on this work and discusses similar work in the literature. Section 3 presents the dataset in detail while Sects. 4 and 5 discuss the ML methods and the experimental result respectively. Section 6 draws conclusion of this work and open issues.

1.1 Motivations

The problem of reconstruction and restoration of little fragments of pictures and frescoes is a very sensitive issue for archaeologists and art historians (Tammisto et al. 2008). Historical dating is an inherently difficult task, but it can be especially useful when faced with the challenges of reconstructing a collapsed wall or recovering the original appearance of a fresco that has been destroyed over time and whose fragments have become mixed with others (Brown et al. 2008). In several cases, the historical dating of an artefact cannot be determined with certainty due to the absence of accurate reproduction of the original, such as a picture or video recording. Frequently, artworks are extracted in extreme conditions and are often damaged or smashed to pieces (Dell'Unto et al. 2013). In this scenario, the historical dating of the artwork becomes uncertain, especially when the recovered fragments are such to prevent the faithful reconstruction of the original aspect of the artwork (Díaz-Marín et al. 2015). Approaching the problem in a traditional way, difficulties in applications and solutions emerge. Due to these factors, having a system that provides information on the style could be especially beneficial during the phase of separating the various fragments, which precedes the reconstruction. The stylistic classification of fragments, the subject of our study, is, to the best of our knowledge, a novel study that we address by implementing both Machine Learning (ML) and Deep Learning (DL) techniques. In fact, although in the literature (see Sect. 2) when implementing a "whole" artwork recognition system it is also done through DL strategies, fragments contain substantially less information than the entire artwork, hence it is probable that DL methods perform less well than anticipated in the overall case. In a preliminary investigation, we evaluated and analyzed both conventional ML and DL classifiers.

1.2 Challenges

The main challenges we faced in this work can be summarized as follows:

- Having data that reflected a possible fragmentation scenario, with fragments thus not necessarily containing semantic information about the artistic content.
- Given the small size of the images and thus, consequently, of the information contained, choose the best strategy with which to extract features and then classify the artistic style.
- Optimize the performance of the chosen models.

1.3 Contributions

The main contributions of the paper can be summarized as follows:

- We propose a dataset called CLEOPATRA (CLassification of frEscOes PictoriAl sTyle fRom frAgments) that simulates the working conditions of archaeologists. Eleven distinct art historical periods are represented, and for each period, at least 40 images were gathered and

split using two distinct fragmentation procedures. This is the first public dataset containing pictures of artworks fragmented in two different ways, constructed with the intention of classifying them stylistically.
- This study demonstrates how tough the stylistic classification task can be, even for the most well-known and effective ML and DL techniques: ML algorithms fail to generalize the problem, especially as the classes considered increase, while DL approaches reveal their limitations on the problem under consideration.
- Given the novelty of the subject of fresco fragment stylistic classification, we believe that our work can serve as a valuable starting point for further research.

2 Related works

2.1 Painting style detection

Every archaeologist is aware of how time-consuming and careful it is to rebuild an artifact (Demetrescu 2015). The same is true for dating a work to a certain historical or artistic era. Every reconstruction is the product of an examination of the ruins and the similarities among the pieces, but it is motivated by experience and occasionally assisted by a little randomness. According to philological reasoning, the attempt to restore the work's original appearance is still within the realm of possibility (Brandi 1963). Because of this, the artistic world is depending more and more on automatic computer analysis to identify and classifier artistic artwork (Limoncelli 2012). Due to the variety and quantity of the existing artistic works, measuring the information contained in them requires effective, and automatic computing. To accomplish this, the model must generate a compressed image representation. The difficulty of identifying the pictorial style of a whole work of art has been studied and analysed for years. In particular, the features of artistic paintings have been analysed using statistical methods and properties such as colour, texture and many others (Bae and Durand 2004; Nunez-Garcia et al. 2018; Sethi et al. 2018). In Nunez-Garcia et al. (2018) the authors studied 7 different artistic period which are: Abstract Expressionism, Cubism, Impressionism, Pop art, Renaissance, Romanticism, and Mexican muralism by the combination of color and texture features, features of color channels from different perceptual color spaces and a subset of seven features selected from the set proposed by Unser (1986) and the feature vector Local Binary Patterns (LBP). Using these features as input, Artificial Neural Networks achieved an accuracy of 70.38 % over 801 photos. Partial damage to a painting obviously affects the accuracy of automatic style recognition, the work (Sandoval et al. 2020) studies this problem implementing a Convolutional Neural Network for the recognition of 20 different fine art categories. The style of the artists, as well as artistic influences and shared techniques, are the focus of the work (Silva et al. 2021). These aspects were evaluated by quantifying the local peculiarities of each painting. For the classification of 13 periods, the constructed model achieved a 65% accuracy rate. A thorough investigation was also conducted by Sandoval et al. (2019). This research presents a novel, two-stage picture classification method designed to increase the accuracy of style classification. Initially, the proposed method separates the input image into five patches and trains and classifies each patch using a deep CNN. In the second stage, the results from each of the five patches are combined in the decision-making module, which employs a shallow neural network trained on the probability vectors provided by the first-stage classifier. The experiment is performed on three datasets containing 6,22,19 different classes. The dataset 3 yields very intriguing results, with a performance of approximately 75%.

2.2 Artistic fragments for classification and reconstruction

The majority of works on style and genre classification are paintings. However, when disruptive conditions occur, very minute fragments, unusual shapes, color changes, and missing parts are just some of the potential issues that can arise (Tammisto et al. 2008). In earthquakes, for instance, a wall painting may crumble, yet the wall holding it may remain standing (Doumas 1992). The organisers of the DAFNE competition (Dondi et al. 2020) investigated the issue of digital anastylosis through a challenge in which participants were challenged to reconstruct a fragmented image (moreover famous frescoes from the Italian art). The task should be based on the fragmented image, and the participants should be able to restore each fragment to its original position. The best performing solutions (Lermé et al. 2020; BAR 2020; Dimov 2020) have demonstrated that classical image processing techniques are superior to machine/deep learning approaches for addressing issues such as digital anastylosis. The DAFNE challenge is simply one of numerous attempts to address the digital image reconstruction problem using computer science approaches. Funkhouser et al. (2011) in their work studied the possibility of employing machine learning approaches to develop a classifier that incorporates numerous fragment-matching properties in order to improve the accuracy of predictions in a fresco reconstruction system. They evaluated matches in three distinct sets of fresco fragments and discovered that classifiers based on several match properties are substantially more effective at scoring suggested matches than scores based on a single match feature. Such a task can be made even harder by the fact that the representation of the data is sparse, due to missing fragments or corrupted data. Inpainting techniques (Rojas

et al. 2020) as well as models working with sparse matrices (Wu et al. 2022; Yuan et al. 2022) can be useful to deal with incomplete data. To classify the 3D Terracotta warriors parts, the authors of (Yang et al. 2020) provide a new classification framework. The proposed framework is built on a dual-modal neural network that can take into account both the spatial location and the texture of the pieces to determine the category for each one. The combination of the two characteristics allows the system to outperform approaches that rely solely on one or the other. Reconstruction is also the goal of work (Du et al. 2016). This research focuses on investigating ancient pottery fragments. They are often found at archaeological sites in a broken state that requires hard work by experienced archaeologists. This work presents the difficulty of reconstructing objects without prior knowledge of the component that should initiate assembly. In order to accomplish this, the authors rely on an earlier phase of classification that allows them to organize the pieces into groups. Then, a neural network determines the component that matches a pair of fragments. In this research, an unsupervised classification is used, whereas in Cascone et al. (2022), the authors addressed a supervised classification issue by considering a working environment with fragments of mixed fresco. This is a crucial preliminary step in the reconstruction of fragmented frescoes, especially when hundreds of small fragments must be distinguished. It is not uncommon to find various pictorial styles in the same room or house: in Pompeii and many other ancient cities, the walls of the dwellings were decorated with a combination of styles (Bragantini 2008; Lepore 2019). In this study, fragments were evaluated with two distinct specific goals: to establish a period classification and an author classification. Various machine learning and deep learning algorithms were compared in an effort to identify the most effective solutions, so providing a useful baseline for researchers working in the field.

3 Dataset

The CLEOPATRA dataset[1] proposed in this study consists in a collection of pieces extracted from paintings focusing on many different art historical periods, which are very different for lines, ways to paints, effects, very easily speaking more recognisable from humans' eyes and for an algorithm. The dataset is organized in eleven classes. It includes artistic periods from the antiquities: Prehistorical, ancient Egyptian, Etruscan, ancient Greek, Roman and Late Antiquities. They are perfectly near in time but very distinguishable each other. The others are: Romanic-Gothic, Renaissance, Impressionism, Cubism, Surrealism. The differences between each period are to explain separating the artistic periods into two groups. In the first group, the antiquities, the Prehistoric style is particularity distinguishable for the essential lines and the elementary subjects. The examples come from the Lascaux cave, the Altamira cave, from the Tassili n'Ajjaer in Sahara Desert, from Ennedi (Ciad), etc. The ancient Egyptian file contains also very recognisable pictures for the profile of people and the use of the colours on the wall paintings. Between the Etruscan and the Greek art the differences are less evident, thus representing a subset of particular interest for classification tasks. For the Roman and Late Antiquity-Byzanthine part, the specific characteristics of the period as the 'megalografia' and the mosaics make the classification task easier.

The second group includes the Romanesque and Gothic historic–artistic period in which are included some frescoes by Beato Angelico, Simone Martini, Giotto. They are characterised by a limited use of colours and the lack of expressions and perspective. The Renaissance art occupies a privileged space for magnificence and uncountable peculiar elements for which is well know: the perfect drawing and the full perspective. The CLEOPATRA dataset collects known examples like Leonardo, Perugino, Botticelli, Raffaello and Giulio Romano. Impressionism is highly recognisable for the peculiar characteristic of the evanescence of the drawing and the privileged use of colours and contrasts of them. The dataset includes in this class very famous painter as Monet, Manet, Degas and Van Gogh. The last two classes are Cubism and Surrealism, which are easy to be recognised from the other classes as well as distinguished by each other when the full parting is available. On the other hand, however, they pose a critical issue when the classification has to be performed on fragments of the artworks, since intra-inter classification becomes hard even for expert in just these two classes. Figure 1 shows a sample per class of the paintings in CLEOPATRA dataset.

3.1 Fragmentation of the dataset

To replicate the disruptive phenomena that may lead a fresco collapsing and falling into pieces, the first part of the work focused on a reliable fragmentation of the images in the dataset. The fragmentation took place following two different approaches, (i) a completely random reduction into fragments, i.e. without taking into account the semantic content of the work and the information that could be contained in each piece and (ii) a salience-based one, which takes into account the salience of the visual stimuli depicted. For the random approach the simple linear iterative clustering algorithm (SLIC) was selected while, for the salience-based fragmentation, the Scale-invariant feature transform (SIFT)

[1] The datasets generated during the current study are available from the corresponding author on reasonable request.

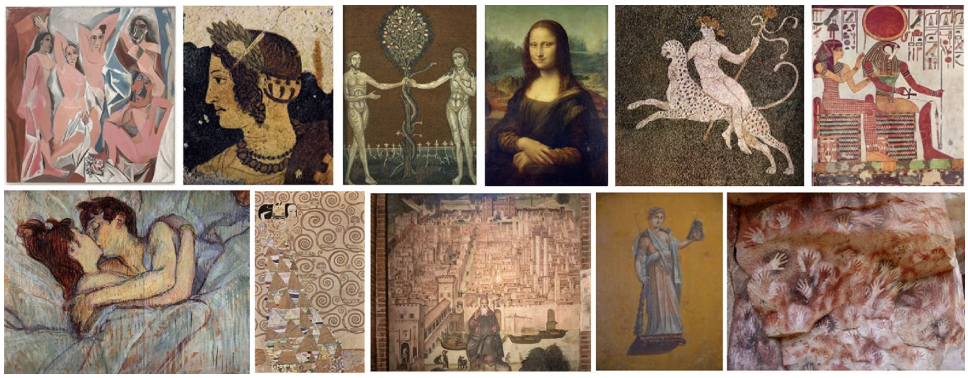


Fig. 1 Examples of the chosen paintings representing the 11 selected artistic periods. From top to bottom, from left to right, respectively Cubism, Etruscan, Late ancient Byzantine, Renaissance, Greek, Egyptian, Impressionism, Surrealism, Gothic, Roman, Prehistoric

operator was chosen. The salience-based fragmentation process continued with the grouping of the extracted features (i.e. a set of points) into clusters, this was possible thanks to the help of the K-means algorithm (Anzalone et al. 2019). Once the clusters have been defined, their contours must be defined. To obtain these fragments contours, for each group the smallest convex polygon is defined which contains, within it, all the points of which each group is formed. A polygon constructed in this way takes the name of convex hull. At the end of both fragmentation strategies, a mask is applied to isolate the resulting fragments. To analyze the impact of the size of the pieces, and therefore also of the amount of information contained in it, datasets were created for each strategy with a variable number of fragments. We started with a split of 10, gradually doubling them down to 80. In total, 4 datasets with clusters of 10, 20, 40 and 80 pieces respectively were created. Figure 2 shows examples of the pieces obtained for the artworks of the Etruscan period both for the salience-based and random fragmentation compared to a grouping of 10, 20, 40 and 80 clusters.

However, the SLIC algorithm does not fragment into an exact number of fragments, in fact the value given in input is indicative and subject to variation for each image. In order to reach approximately the desired number of fragments, where it did not happen automatically, the value that regulates the proximity of the color and the proximity of the space has been adapted from time to time. We therefore varied the compactness value of a superpixel that balances the proximity of the color space with the proximity of the image space. Since the greater this value, the more compact and square the cluster will be, we tend to set it low but, if it is not possible to reach the target number of fragments, we progressively increase it each time from 5 to 40 with a spread of 5 and from 40 to 46 with a spread of 2. If setting this value to 46 still does not reach the desired number of fragments, in any case, we stop to avoid having shapes that are too square and therefore unrealistic. After having obtained both fragmentation, we proceeded to eliminate the background leaving in fact only the visible fragments. Subsequently, all those pieces were manually removed for which, for example, irrelevant information was contained, such as pieces depicting only walls, cave walls, etc. This operation was mainly carried out on the folders relating to Egyptian, Etruscan, Greek, Prehistoric, Roman and Byzantine art.

3.2 Dataset organization

After having fragmented the selected paintings as explained in the previous Sect. 3.1, we proceeded with the relative organization. For each type of approach, both the conscious and the random one, folders related to the 4 types of fragmentation 10, 20, 40 and 80 have been created. With respect to each approach and with respect to each fragmentation strategy, 3 folders have been created, that of train, test and validation. This subdivision was made in such a way that among the various folders there are never fragments belonging to the same painting. The choice was made upstream on the paintings. Therefore, 26 artworks were selected for the train subset, 6 for validation one and, finally, 8 for the test one. The motivation behind this choice was therefore to ensure that a ML algorithm does not catalog the fragment because it has already been seen in some way, but because it can instead capture those aspects that allow it to be classified as belonging to one period or another.



Fig. 2 From left to right, examples of the pieces obtained for the artworks of the Etruscan period both for the random (upper) and salience-based (bottom) fragmentation compared to a grouping of 10, 20, 40 and 80 clusters

4 Materials and methods

4.1 SLIC

SLIC is a fast and rather memory efficient algorithm that allows you to group pixels that share common characteristics (Achanta et al. 2012). These sets of pixels are called superpixels and can give a compact representation of the image. These groupings are performed in a 5-dimensional space based on the position, i.e. x and y coordinates of the pixels, and on the three values L^*, a^*, b^* of the CIELAB color space. L^* refers to the lightness value while a^* and b^* are the chromatic coordinates that identify the hue and saturation. The algorithm requires only one parameter, a certain k which represents the number of superpixels of approximately equal size to obtain. Hence, this algorithm uses soft k-means clustering to generate these pixel groups. The basic idea is to generate local clusters in iterative evolution until obtaining what are then the final groupings. In the initial phase, a regular grid S of the centers $C_i = [l_i, a_i, b_i, x_i, y_i]^T$ of the clusters is defined on the image, i.e. they are placed equidistant. Thus, if n is the number of pixels, each cluster has an approximate size of the type n/k. In the next steps the centers are moved to seed locations on the basis the lowest gradient position in a 3×3 neighborhood. This ensures that the center of the clusters does not end up on an edge and reduces the possibility of choosing a noisy pixel (Achanta et al. 2010). The gradients are calculated as:

$$G(x,y) = \|I(x+1,y) - I(x-1,y)\|^2 + \|I(x,y+1) - I(x,y-1)\|^2 \quad (1)$$

where $I(x, y)$ is the CIELAB vector of the pixel at the position x, y. After all the pixels are linked to the center of the closest cluster taking into consideration only those whose search area contained them, a new center is defined by calculating the average of the vectors representative of the pixels of the same cluster. The distance function behind the concept of proximity takes into account the fact that if the distance is limited for the color space, this does not happen for the spatial components linked to the size of the images. So it is necessary to adopt some kind of normalization:

$$D = \sqrt{d_c^2 + \left(\frac{d_s}{S}\right)^2 m^2} \quad (2)$$

where d_c and d_S are the euclidean distances respect the CIELAB coordinates and spacial ones, respectively and m is a weight that determines the importance of color or spatial similarity. The algorithm stops when convergence is achieved.

4.2 SIFT

The Scale-Invariant Feature Transform (SIFT) algorithm is a technique for extracting local features that are invariant with respect to basic image transformations such as scale, translation and rotation, but also stable with respect to changes in illumination and noise. Besides this, an additional advantage is the large number of features that can usually be extracted for each image.The algorithm is composed of 4 main steps: Scale-spase extrama detection, Keypoint localization, Orientation assignment and the Keypoint descriptor (Purandare and Talele 2014).

1. *Scale-spase extrema detection*. The spatial scale aims to replicate the concept of multiscale in digital images. In this first step, therefore, the method detects stable features in all possible scale spaces. The scale space of an image I is defined as a function $L(x, y, \sigma)$ which is obtained from the convolution between the image and a variable-scale Gaussian kernel $G(x, y, \sigma)$:

$$L(x,y,\sigma) = G(x,y,\sigma) \cdot I(x,y) \quad (3)$$

where (x, y) are the coordinates of the image I and σ is the scale factor, σ is set as optimal value to 1.6. The Gaussian kernel is defined as:

$$G(x,y,\sigma) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x^2+y^2)}{2\sigma^2}}. \quad (4)$$

The scale-space is divided into octaves, each octave being an image with half the size of the previous one. The number of octaves depends on the size of the original image. A so-called Gaussian pyramid is therefore generated from the original images, where each level is therefore an octave. To discover interesting keypoints in the image are used difference of Gaussian (DoG) images. This type of image is generated by subtracting two neighbouring scales that are separated by a certain constant multiplicative factor k set to $\sqrt{2}$:

$$D(x,y,\sigma) = L(x,y,k\sigma) - L(x,y,\sigma) = (G(x,y,k\sigma) - G(x,y,k\sigma)) \cdot I(x,y) \quad (5)$$

This process is then performed on different octaves of the image in the Gaussian pyramid. After getting the whole DoG pyramid constructed, the most interesting features from the DoG images are selected. To detect local extremes, each pixel is compared with its neighbours in a 3×3 region and nine neighbours in adjacent scales. Each candidate pixel is selected only if it has a greater or lesser value than all its neighbours. A local extreme, is potentially a keypoint.
2. *Keypoint localization*. This is the step in determining position and scale of the keypoints. In the previous step

many keypoints are generated, some of which, however, are not useful and must be discarded because they have low contrast or are located along an edge. To remove points that do not have enough contrast, Taylor series expansion is used:

$$D(x) = D + \frac{\partial D^T}{\partial x}x + \frac{1}{2}x^T\frac{\partial^2 D}{\partial x^2}x \tag{6}$$

where D and its derivatives are evaluated at the point under consideration. If the intensity of the point under examination is less than that of a certain threshold then it is removed. The DoG function has a high response along the edges. However, for a keypoint the position along the edges is poorly determined as it would not be robust to small amounts of noise. Therefore, there is a need to purge these values by exploiting the fact that these types of points have a larger principal curvature along the edge than in the perpendicular direction. These curvatures can be calculated by means of a 2×2 Hessian matrix H of the second-order:

$$H = \begin{bmatrix} Dxx & Dxy \\ Dyx & Dyy \end{bmatrix}. \tag{7}$$

3. *Orientation assignment*. After determining the scale in which the keypoint is detected, we move on to assign an orientation to each of them to make them invariant to rotation. Information from the image of the keypoint's neighborhood is used to calculate these directions. Therefore, the value of the module $m(x, y)$ and the orientation $\theta(x, y)$ for each sample $L(x, y)$ are calculated as follows:

$$m(x, y) = \sqrt{l_x(x, y)^2 + l_y(x, y)^2} \tag{8}$$

$$\theta(x, y) = arctan\left(\frac{l_y(x, y)}{l_x(x, y)}\right) \tag{9}$$

where $l_x(x, y) = L(x + 1, y) - L(x - 1, y)$ and $l_y(x, y) = L(x, y + 1) - L(x, y - 1)$. Once the modulus and angles are calculated, the orientation histogram is created. This consists of the gradient of sample points within a region around the keypoint. The orientation histogram has 36 bins which therefore cover the 360° orientation range with an amplitude of 10° for each. A sample before being added to the histogram is weighted by its gradient magnitude $m(x, y)$ and by a Gaussian weighted circular window with a σ that is set equal to 1.5 times that of the keypoint scale. At the end of this operation repeated for all pixels around the keypoint, the histogram will have peaks. These peaks are indicative of the dominant direction of the local gradients.

4. *Keypoint descriptor*. In this last point a descriptor, a normalized 128-dimensional vector, is created for each keypoint. At this stage, in fact, each keypoint has a position, a scale and an orientation. A keypoint descriptor is created for each point in the 16×16 keypoint neighbourhood image. Then, each window is divided into 4×4 sub-regions, resulting in 4 different areas, with 8 orientation bins in each. This gives a feature vector of $4 \times 4 \times 8 = 128$ elements for each keypoint. To reduce the effects of lighting change, it is normalised to unit length.

4.3 Classification of the fragments

In the second phase, for the classification of the various fragments, different types of ML algorithms were implemented, each with its own peculiarities. Two different approaches were evaluated. In the first strategy, neural networks were used with input images of the fragments themselves; in the second, however, descriptive information was extracted from each fragment using known algorithms and then supplied as input to three different classifiers. Several models of Convolutional Neural Networks, AlexNet, ResNet-50, ResNeXt-50, and a custom CNN model were tested. The most famous and used classifiers have been selected, namely Random Forest, SVM and KNN. If the networks take images as input, this does not happen for the classifiers, which instead require numerical data. Therefore, 3 descriptors of the qualitative properties of the images were selected, i.e. characteristics of color, texture and shape that allow to characterize their content. For color information, color histograms are extracted. They are a representation of the color distribution for each of the R, G and B channels. The Hu moments are, instead, shape descriptors, particularly used because they are invariant to translations, rotations and scaling. Shape element vectors are extracted from the silhouette or outline of an element in an image. Basically, the moments of the image are defined as the weighted average of the pixel intensities. Finally, Haralick texture vectors capture the information that emerges in texture patterns. Underlying this process, the Gray Level Co-Occurrence Matrix (GLCM) is used to calculate 13 characteristics. It takes advantage of the adjacency concept in the images. The basic idea is to look for the relationship between two adjacent pixels at a time to look for different combinations of pixel brightness values. This information, extracted for each fragment, obtained from the color histogram, the Hu moments and the Haralick texture vectors was concatenated, scaled and translated in a range of (0,1) and then given as input to the classifiers. Since these require the setting of several parameters, several were selected and then, through an exhaustive search, the best combination was found, the one that would have returned the highest accuracy. In addition, a k-fold cross-validation

Table 1 Comparative analysis between the three selected algorithms in terms of accuracy

Dataset	A_10			A_20			A_40			A_80		
Classifier	SVM	KNN	RF	SVM	KNN	RF	SVM	KNN	RF	SVM	KNN	RF
F1	0.2473	0.21	**0.3319**	0.27	0.20	**0.34**	0.24	0.21	**0.32**	0.18	0.19	**0.28**
Precision	0.33	0.22	**0.34**	0.30	0.20	**0.35**	0.27	0.22	**0.32**	0.28	0.19	**0.29**
Accuracy	0.26	0.22	**0.34**	0.29	0.20	**0.34**	0.26	0.22	**0.33**	0.22	0.19	**0.30**
k	6	2	4	2	2	6	4	2	2	2	2	9
	R_10			R_20			R_40			R_80		
F1	0.2576	0.20	**0.3446**	0.25	0.18	**0.31**	0.20	0.19	**0.3085**	0.16	0.18	**0.27**
Precision	0.28	0.24	**0.38**	0.29	0.17	**0.34**	0.31	0.19	**0.33**	0.18	0.19	**0.29**
Accuracy	0.26	0.22	**0.34**	0.28	0.18	**0.31**	0.21	0.19	**0.31**	0.18	0.19	**0.28**
k	2	2	10	2	2	4	2	2	4	2	5	4

The results presented are the best, obtained after doing an exhaustive search of their hyperparameters. In bold, the best performances are highlighted

procedure was also used with a variable " k " between 2 and 10. This technique allows you to divide the dataset training into k parts and train the model on a subset consisting only of k-1. The remaining part is used to validate the model. The measure of performance obtained is the average of the individually derived k-1 values.

5 Experimental results and discussion

First Strategy The experimental analysis was carried out on the datasets obtained as explained in Sect. 3.2. For the first experimental approach it was decided to work directly on the images and then to use a series of Convolutional Neural Networks to obtain the desired classification. The models were trained on the train set, validated on the validation test and finally tested on the test set. The results achieved were far below expectations, never managing to exceed an accuracy value of 10% in many cases, which is why we chose not to continue with this strategy but rather try a completely different one. In fact, this type of approach seems not to be adequate to extrapolate from the fragments, regardless of their size, the adequate information to obtain a good classification.

Second Strategy We decided to implement a more analytical approach, that is, extracting information from each fragment of shape, color and texture and using it as input for three classifiers: Random Forest, KNN and SVM. In this case the models were trained on the merger of the two train and validation datasets and then tested on the test set. As mentioned in the previous section, several parameters were selected and then through an exhaustive search those in correspondence with which the highest accuracy was obtained were selected. The results obtained are shown in Table 1. It is evident that the classifier that seems to give the strongest response, with respect to all fragmentation strategies, is Random Forest, followed by SVM and then finally KNN. The difference between these three models for each experiment is clear. On average Random Forest is superior to SVM with an accuracy percentage of about 8%, with trials where it reaches 10%, while the average difference between SVM and KNN is lower, at 5%. Moreover, as expected, with the same number of image fragments, the conscious strategy allows to obtain greater results, even if the difference is not as clear as one might hypothesise, on average only about 2% is the improvement recorded. The sharpest difference between the two approaches is recorded for fragmentation in 20 pieces, about 4%. Minimal difference for the 10-piece fragmentation, the size of the fragments seems to make up for the random approach and flatten out any semantic differences. Surprisingly, moreover, when varying the size of the fragments the difference does not even seem to be so sharp, within the same approach (conscious or random). In fact, if we compare the results of the fragmentation of 10 pieces with that of 80, for the random strategy there is a difference of 6% while for the salience-based one of about 4%. In fact, it would seem that with this last approach, selecting the fragments on the basis of the characteristics detected by SIFT, the different size of the fragments and therefore, consequently, the lower information contained in them, have less impact on the system performance. Figure 3 illustrates the confusion matrix related to the random experimentation with 80 pieces using the Random Forest approach. This setup, as shown in Table 1, proves to be the most difficult, which is particularly intriguing from our perspective because it more closely resembles a real-world scenario in which a work is fragmented. This graphic depiction lends itself to numerous concerns. The matrix of confusion between Romanic-Gothic and Roman pieces reveals one of the most significant ambiguities of the model, which is not surprising

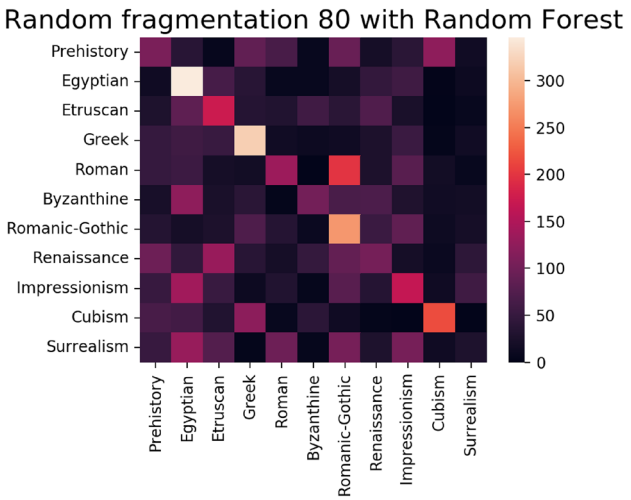


Fig. 3 Confusion matrix for random fragmentation, 80 pieces with Random Forest

given that for the Romanic-Gothic art current, the adjective "Romanic" refers specifically to the Roman matrix of this art form, which developed in the ancient territories conquered by the Romans and which also adopts some ancient techniques. Egyptian artefacts have the highest level of recognition. Again, the model's response is consistent with what may have been expected, given that the pure, bright colours juxtaposed by contrast and the forms delineated by a precise contour render the fragments especially noticeable.

5.1 Ablation study: progressive number of artistic periods

After this first experimentation, it was decided to specifically analyse the most challenging one, the one that, in any case, is most likely to occur in reality, namely the random fragmentation of 80 pieces. Therefore, in this further study, the classification was limited to subsets of artistic periods of progressive size. First the three artistic periods considered most dissimilar to each other were selected, and then two others were gradually added with the same criterion of considering the most dissimilar each time. Given that in the first experiment for each dataset, Random Forest is the classifier with which the highest results are associated, in this new experiment it was decided to apply only this algorithm. The exhaustive search strategy allowed us to obtain the values shown in the Table 2. From the results in the table it can be seen that, by considering the first three most dissimilar artistic periods in terms of shape, colour and texture, and then adding two more with the same criteria, the accuracy drops sharply. In fact, there is a performance difference of 20%, a much smaller difference when going from 5 to 7 periods, from 7 to 9 and then from 9 to 11.

Table 2 Comparative analysis between a progressive number of artistic periods in terms of accuracy

Artistic periods	F1	Precision	Accuracy	k
[3]: Egyptian, Cubism, Renaissance	0.6894	0.6921	0.6912	8
[5]: [3] + Prehistory, Impressionism	0.4815	0.5104	0.4975	2
[7]: [5]+ Greek, Roman	0.3947	0.4235	0.4089	4
[9]: [7] + Byzantine, Surrealism	0.3115	0.3291	0.3180	4
[11]	0.27	0.29	0.28	4

The results are obtained after making an exhaustive search of the hyperparameters of the selected algorithm, that is Random Forest

6 Conclusions

The classification of frescoes pictorial style from fragments is a great challenge for archaeologists and art historians. To such a purpose, 11 artistic periods that cover thousands of years were identified, going from Prehistory to Surrealism. A total of 40 paintings were selected per period, the most significant and representative ones for each. The collection of artworks realised the proposed CLEOPATRA dataset (CLassification of frEs-cOes PictoriAl sTyle fRom frAgments), which rather than being a simple collection of artworks is composed by fragmentation of each artwork with respect to 4 different orders of magnitude: 10, 20, 40 and 80 pieces. To the best of our knowledge, this specific perspective of the problem has not yet been studied in depth in the literature and there are no datasets specifically meant to simulate this working conditions of archaeologists. After collecting the datasets of fragments, next step was the extraction of the visual features from each of them. In particular, we focused on the characteristics of shape, color and texture. This representation was used as input for three different ML algorithms: KNN, SVM and Random Forest. From the experimental results Random Forest confirmed its superiority in all experimental conditions. The highest accuracy, i.e. 34.42%, was reported for the salience-based fragmentation with 20 fragments. On the contrary, the worst performances of the 27.62% system correspond to a random fragmentation at 80 pieces. To investigate on scenarios at increasing level of difficulties, we analyzed analyzed the difference among three artistic periods that, on the type of information extracted, resulted the most different and recognizable from each other. Progressively, we increased the difficulties of the problem by adding 2 pictorial style per experiments up to the entire CLEOPATRA dataset has been considered. The results achieved in this setting shows that when the problem is significantly relaxed the classification task is easier and feasible, achieving about 70% in accuracy. The problems becomes rapidly difficult by working on five classes turning into a true challenging task when all pictorial styles must be

classified. The contribution of the achieved results suggests that Machine Learning solutions could not be effective as a solution to the classification problem in this study. Preliminary experiments not discussed in this work, show that deep learning models could also not an ideal choice for the nature of the problem. Ad-hoc image processing techniques for the extraction of low-level visual features can represent the fruitful strategy to improve the classification accuracy. Further improvement of this work may therefore focus on the selection and processing of more discriminating characteristics as well as expanding the CLEOPATRA dataset incorporating artistic movements and variants. In addition, it would also be particularly interesting to investigate in the future the presence of noise in the original frescoes as well as the study of the boundary dimension of the fragments. These new assumptions would therefore lead to further expansion of the dataset.

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